BINARY CLASSIFICATION • HEALTHCARE ML • CAPSTONE PROJECT

Predicting Diabetes Diagnosis

from Lifestyle and Clinical Screening Data

Can routine screening data — demographics, lifestyle habits, and basic clinical measurements — flag who is likely to have diabetes, before a formal diagnostic workup?

100,000	31	5	0.945
Patient Records	Features	Models Compared	Best ROC-AUC

The Problem

Diabetes is frequently under-diagnosed until complications appear

Objective

Build a model that flags high-risk individuals from routine screening data — the kind collected in an annual check-up — to support earlier referral for formal diagnostic testing.

This is a triage / pre-screening aid — not a replacement for clinical diagnosis.

TYPE	Binary Classification
TARGET	diagnosed_diabetes (1 = diagnosed, 0 = not)
UNIT OF ANALYSIS	One individual per row
SUCCESS LOOKS LIKE	High Recall on the positive class — catching true cases matters more than avoiding a false alarm

Why Machine Learning?

- No single clinical cutoff separates the classes once demographic and lifestyle factors are mixed in
- Many weak signals can be combined into one risk estimate
- The model can be re-evaluated as new screening data arrives

The Dataset

100,000 individuals — demographic, lifestyle, and clinical screening measurements

100,000	31	60 / 40	0%
Rows	Columns	Class Split	Missing Values

Three Feature Groups

D

Demographic & Lifestyle

age, gender, ethnicity, education, income, employment, smoking, alcohol, activity, diet, sleep, screen time

R

Personal & Family Risk History

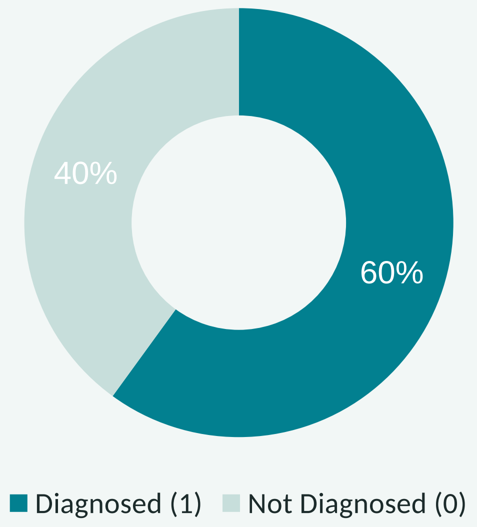
family history of diabetes, hypertension history, cardiovascular history

C

Clinical Measurements

BMI, waist-to-hip ratio, blood pressure, cholesterol panel, glucose (fasting & post-meal), insulin, HbA1c, risk score

Target Distribution



Data Quality & Cleaning

A clean dataset — but two decisions still mattered a great deal

0	0	154
Missing Values	Duplicate Rows	Impossible BP Rows Dropped

Invalid Value Check

systolic_bp ≤ diastolic_bp is physiologically impossible — systolic pressure is always higher in a living person.

154 rows (0.15%) affected — dropped as unambiguous data-entry errors. No safe way to correct a swapped pair, and the fraction is too small to bias the class balance.

Leakage Check: diabetes_stage

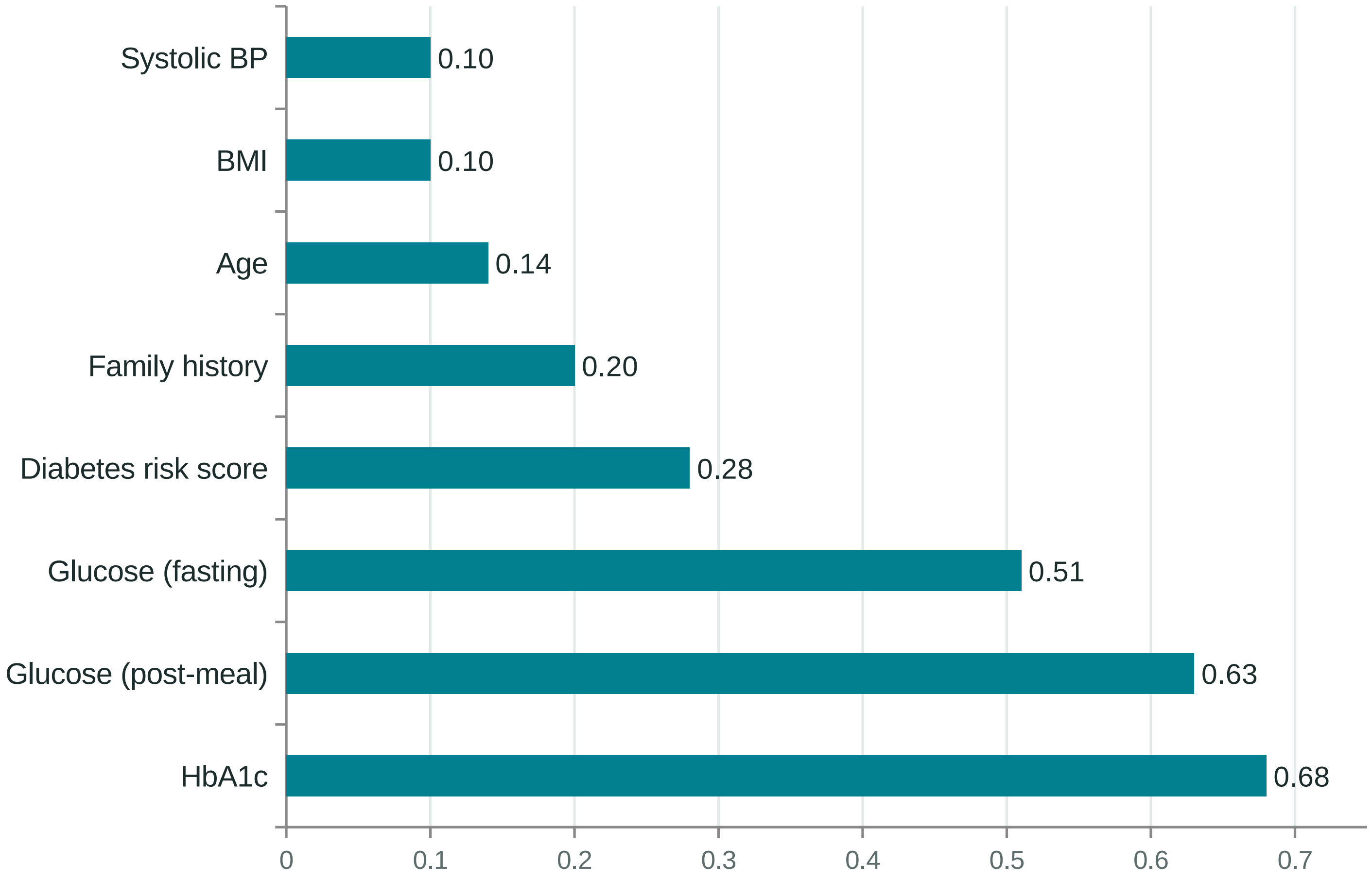
Pre-Diabetes → 100% maps to class 0

Type 2 → 100% maps to class 1

This column restates the diagnosis itself. Keeping it would give near-perfect validation scores that collapse in real use — the clinical stage doesn't exist yet when a screening prediction is actually needed. Dropped from the feature set entirely.

What Actually Predicts a Diagnosis?

Correlation of each feature with diagnosed_diabetes



Reading the ranking

- The top 3 features are the actual clinical diagnostic criteria for diabetes — expected, and a good sanity check
- diabetes_risk_score correlates only moderately (0.28) — not a leaked copy of the target, kept as a legitimate feature
- Lifestyle self-reports (diet, sleep, screen time) show near-zero correlation individually
- No dangerous multicollinearity found among features (outside the glucose/HbA1c cluster itself)

From Raw Data to Model-Ready Features

Every step fit on the training set only — no leakage into validation or test

1

Split First

60% train / 20% val / 20% test
Stratified on the target

→

2

Encode + Scale

One-Hot for categoricals
StandardScaler for numerics

→

3

Variance Filter

Drop near-constant
one-hot columns

→

4

Fit on Train Only

Transform-only on
val / test sets

50 features → 20 numeric (scaled) + 30 one-hot categorical columns after encoding

Two safeguards against leakage: the train/val/test split happens before any preprocessing is fit, and every transformer (encoder, scaler, variance filter) is fit on the training set only.

Five Models, Four Families

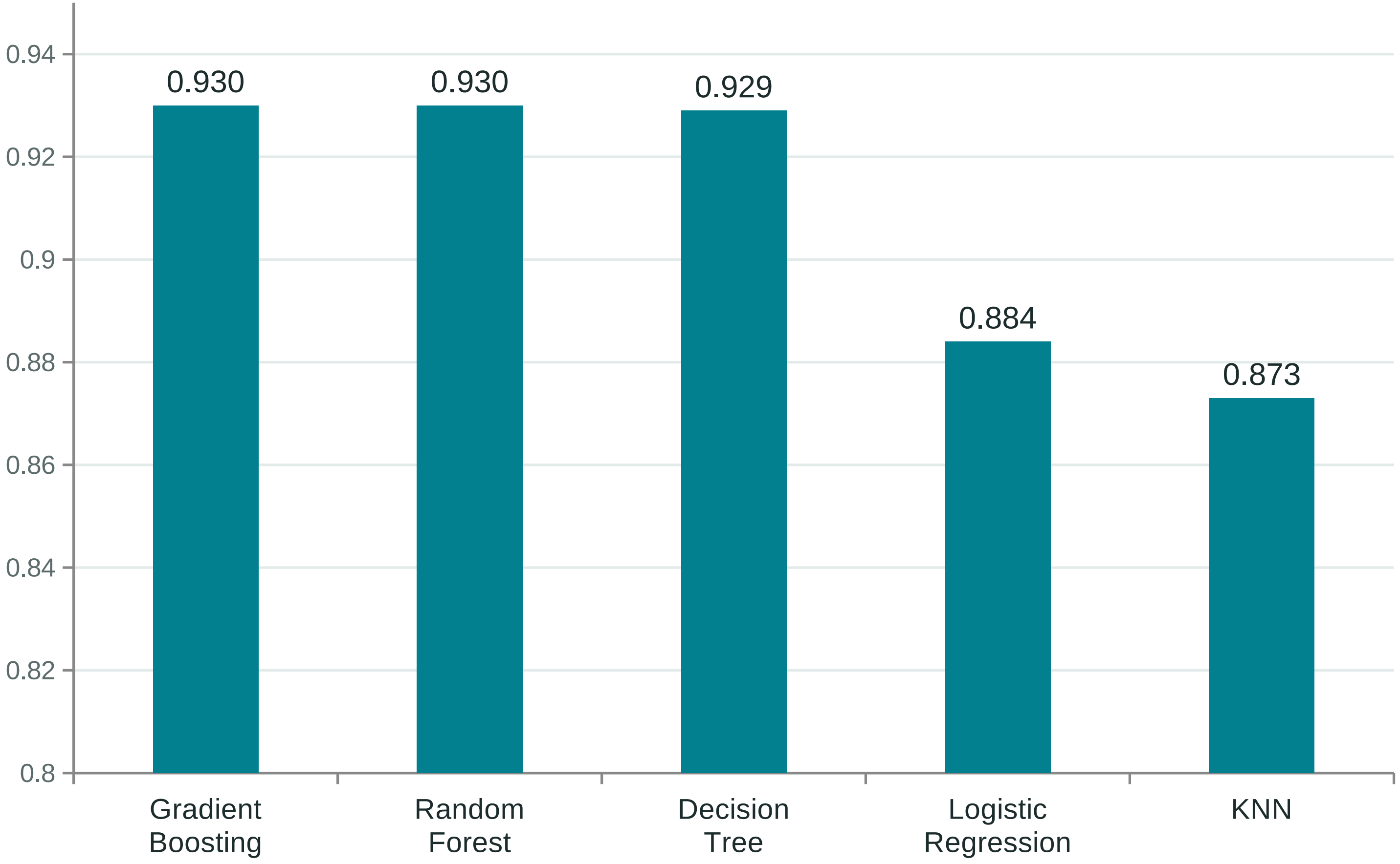
Linear, instance-based, single-tree, and two ensemble approaches

LR	KNN	DT	RF	GB
Logistic Regression	**K-Nearest Neighbors**	**Decision Tree**	**Random Forest**	**Gradient Boosting**
Linear	Instance-based	Non-linear	Ensemble (bagging)	Ensemble (boosting)
Interpretable baseline — sets the floor every other model must beat	No assumption about decision boundary shape	Fully interpretable; captures feature interactions	Averages many trees to reduce variance / overfitting	Sequentially corrects prior errors — often strongest on tabular data

Every model answers the same 8 design questions: why chosen · linear or non-linear · needs scaling? · what it learns · how it predicts · key hyperparameters · strengths · limitations

Model Comparison — Validation Set

After hyperparameter tuning (Val F1, positive class)



Reading the gap

The top three (Gradient Boosting, Random Forest, Decision Tree) are separated by less than 0.1% — not practically meaningful on their own.

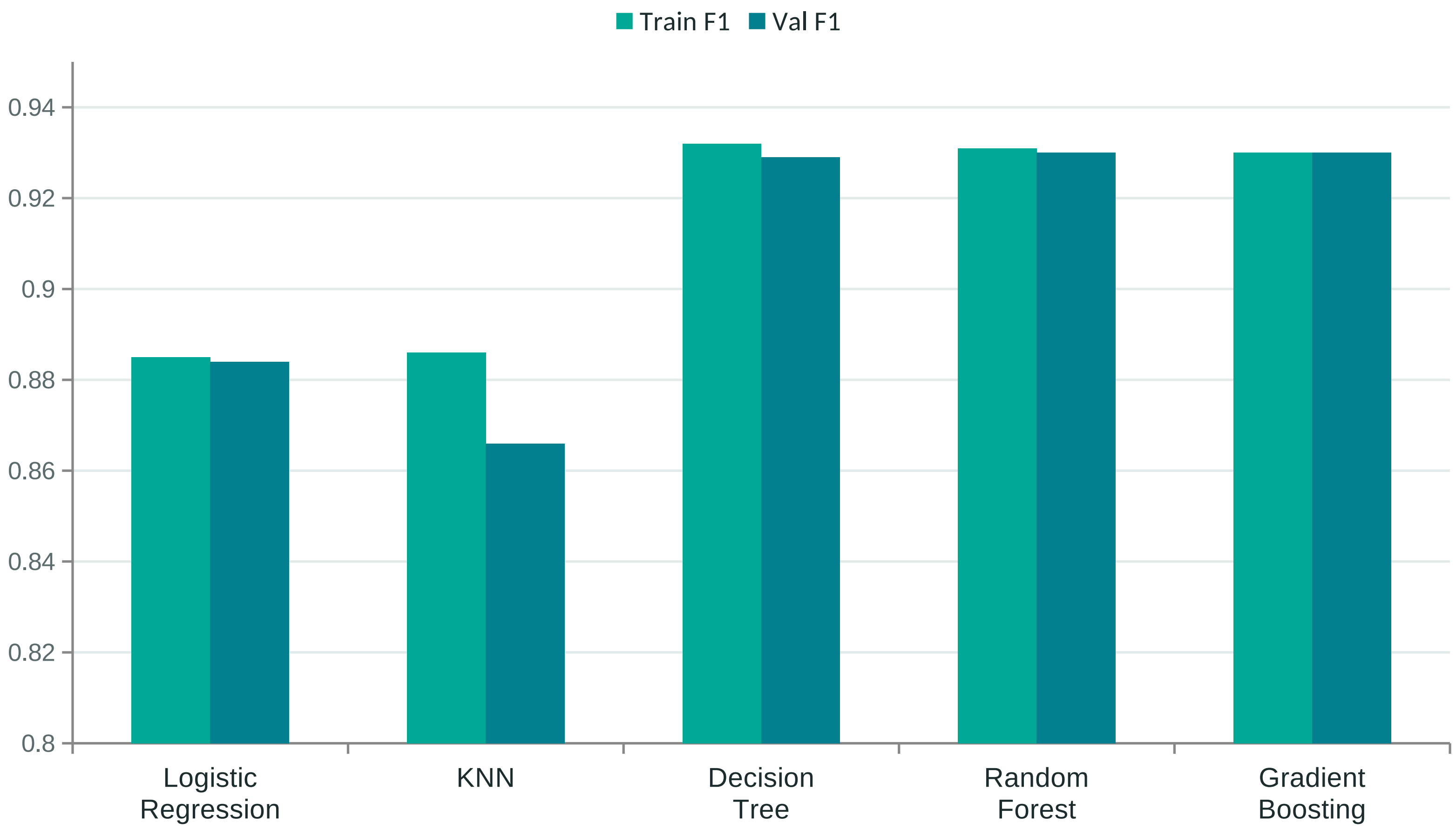
Moving from a linear model to any tree-based model is what drives the real gain — not the specific choice among the top three.

Logistic Regression
ROC-AUC 0.934

Gradient Boosting
ROC-AUC 0.946 (best)

Checking for Overfitting

Train vs. Validation F1 — the gap is the evidence, not the score alone



Largest gap: KNN

2.4-point Accuracy gap — the largest of the five, though still moderate, not severe overfitting.

Smallest gap: Gradient Boosting

Under 0.05 points after tuning — strong generalization, not just a high score.

Tuning (Part 11) narrowed the tree-model gaps that were wider before depth/estimator limits were applied.

Final Model: Gradient Boosting

Selected — then re-tuned for the real-world cost of a missed case

Why Gradient Boosting

Best validation F1 (0.930) and ROC-AUC (0.946), with a train-val gap no larger than its closest competitors — the edge isn't coming from overfitting.

The Honest Caveat

Gradient Boosting leads Random Forest and Decision Tree by under 0.1% F1 — not practically meaningful. Most of the gain over the Logistic Regression baseline comes from using any tree-based model, not this specific choice.

Threshold Tuning — Optimizing for Recall

In a screening tool, a missed diabetic case (False Negative) is costlier than a false alarm (False Positive) — so the default 0.5 cutoff isn't necessarily right.

Threshold	Precision	Recall	F1
0.5 (default)	1.000	0.869	0.930
0.2 (chosen)	**0.863**	**0.909**	**0.885**

0.2

chosen decision threshold

Trades some precision for a meaningful recall gain — fewer real cases slip through

Final Test Set Performance

The one and only look at the test set — after model & threshold were locked in

85.9%	0.886	0.945	91%
Accuracy	F1 Score	ROC-AUC	Recall (class 1)

Confusion Matrix — Test Set (n = 19,970)

	Predicted: No	Predicted: Yes
Actual: No	6,221	1,767
Actual: Yes	1,057	10,925

What this means

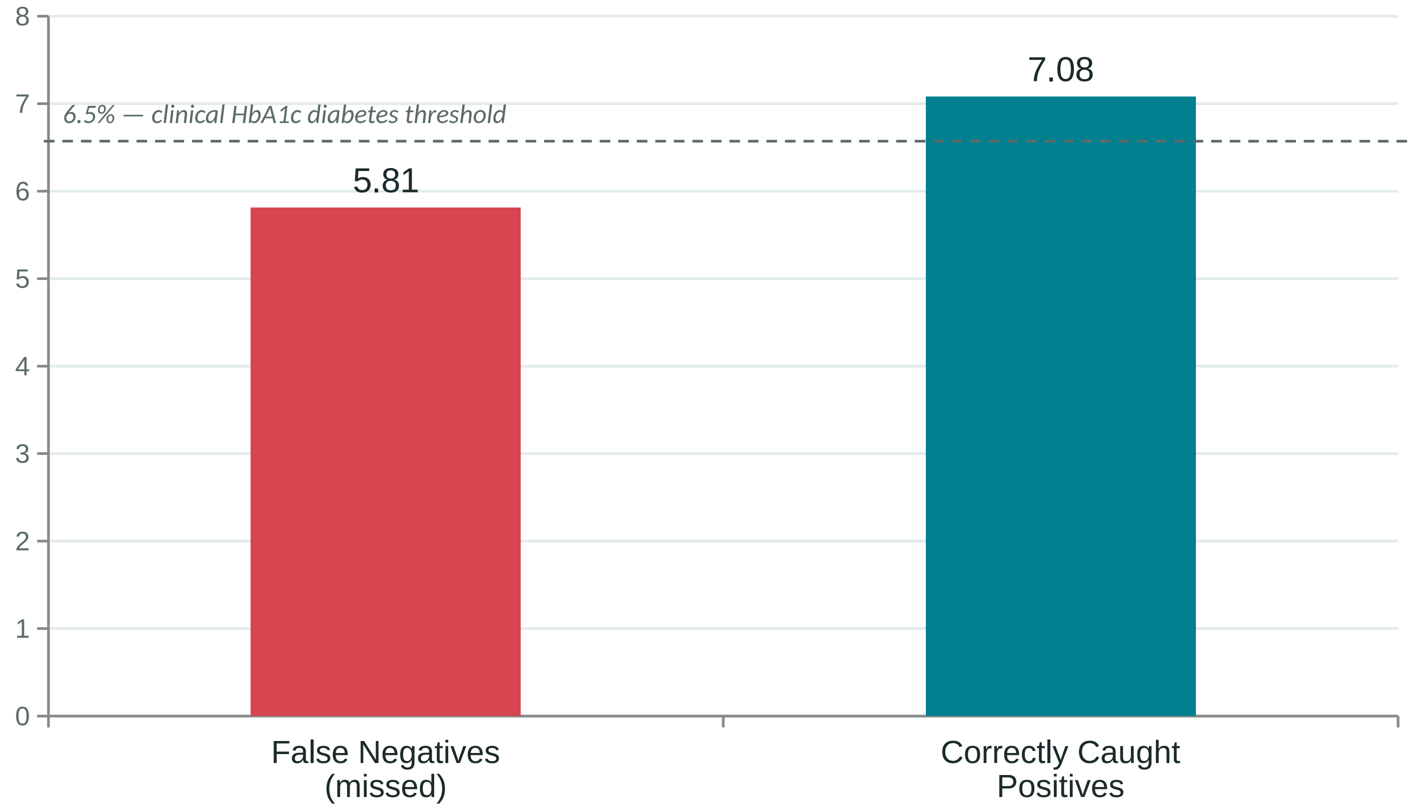
1,057 False Negatives (8.8% of true positives) — real diabetic cases the model misses

1,767 False Positives (22.1% of true negatives) — unnecessary follow-up tests, the accepted cost of prioritizing recall

Where the Model Struggles

False Negatives cluster near the clinical decision boundary — not randomly

Mean HbA1c: Missed Cases vs. Correctly Caught



Three Findings

- Missed cases sit closer to the diagnostic thresholds — the model fails on borderline patients, not obvious ones
- In this context, a missed case is more costly than a false alarm — the reasoning behind weighting Recall in model selection
- Lifestyle/demographic features add limited help resolving genuinely borderline clinical cases

Beyond Classification: Clustering & PCA

Unsupervised structure — and one technique that didn't pay off

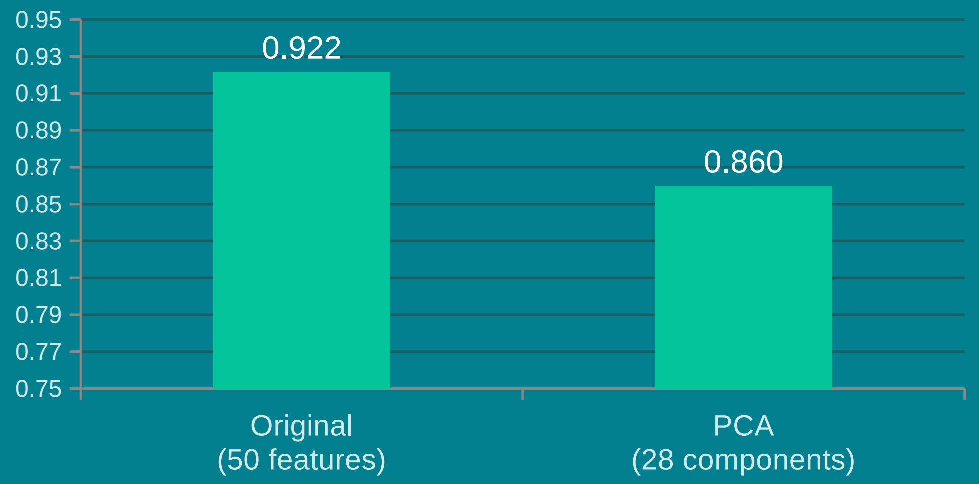
K-Means Clustering (K = 4)

Formed on clinical/lifestyle features only — the target was never used to build the clusters, only to interpret them afterward.

Cluster	Age	Diagnosed Rate
High-risk (older, high HbA1c)	61.5	93%
High-risk (younger, high HbA1c)	48.8	91%
Moderate	51.0	33%
Low-risk / active lifestyle	40.4	26%

Purely unsupervised grouping reconstructed a risk-like gradient that lines up with actual diagnosis rates — a strong sanity check on the feature set.

PCA — Did It Help?



PCA hurt performance: 28 components (95% variance) still dropped Val Accuracy by ~6 points. The target's signal is concentrated in a few high-value features (HbA1c, glucose) — exactly what variance-based compression can discard.

Conclusions & Recommendations

Recommended use: a supportive triage aid for clinicians who already have lab results — not a standalone consumer screening tool that replaces lab testing.

Key Limitations

- Data appears synthetically clean (zero missing/duplicate values) — real deployment data will likely be messier
- Depends heavily on lab values (HbA1c, glucose) that require an actual blood draw
- No external validation on a separate population or cohort

Next Steps

- Test a lifestyle-only feature subset — the more genuinely useful pre-screening product
- Add k-fold cross-validation for more robust hyperparameter selection
- Validate against an external, more diverse clinical population

Thank You